DP Wizard Tutorial

Getting Started with DP Wizard: A Step-by-Step Guide

Overview

DP Wizard is a browser-based tool that helps you build differentially private analyses without needing to write code. It works by walking you through a user-friendly interface to configure the settings for your differentially private release, and it generates a Jupyter notebook that uses the OpenDP Library to construct the actual differentially private results.

This tutorial walks you through how to use the DP Wizard tool on a hosted server.

What You'll Need

- A modern web browser
- A CSV dataset:
    - Private dataset: mock_data_private.csv[1]
    - The dataset contains four columns: first and last name, age (as a number), and city ID (as a number between 1 and 10)
- DP Wizard Server Link [2]
- Notebook Server Link

Goal and Workflow

In this tutorial, we'll use DP Wizard to create a Jupyter Notebook that calculates the **average age of individuals in each city**. In SQL, the corresponding query might be:

SELECT city_id, AVG(age)
FROM data
GROUP BY city_id

To accomplish this task, we'll:

1. Use DP Wizard to create a privacy-preserving analysis and download the Jupyter notebook.
2. Upload the notebook to a Jupyter Notebook server and run it to see the results.

---

[1] Links that could reveal author or institutional identity have been removed from this archived version for anonymous review. The corresponding study materials are provided separately in the anonymous artifact repository.

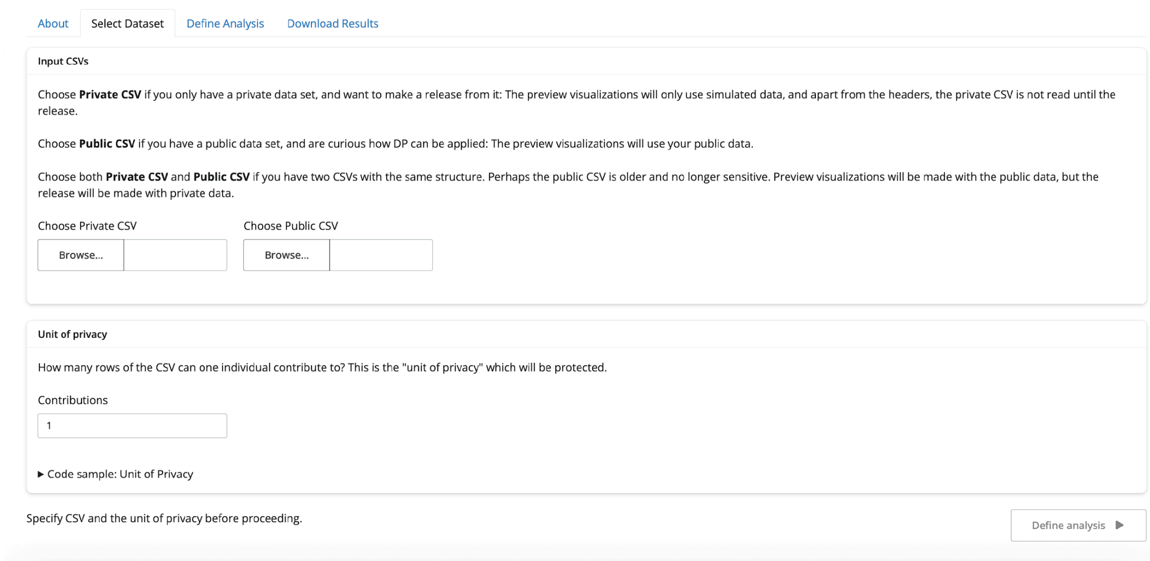
[2] The DP Wizard and Jupyter Notebook servers were temporarily hosted for the study and were taken down after data collection. The original server URLs have been removed from the archived study materials to preserve anonymity.

Step-by-Step Instructions

Step 1: Select Dataset

Open DP Wizard.

The first tab, called “Select Dataset,” allows you to upload your dataset as a CSV file.



Download the dataset linked at the top of the tutorial to your computer, then upload in the “Choose Private CSV” box in DP Wizard (shown in the screenshot above).

The “Unit of privacy” box allows you to specify the unit of privacy protection your results will provide. DP Wizard’s goal is to protect individuals; to do this, it needs to know how many rows of data may belong or correspond to a particular individual. For our dataset of individuals and ages, each individual corresponds to a single row in the dataset, so we can leave the default value of “1” in this box.

Step 2: Define Analysis

The second tab, “Define Analysis,” allows you to specify the query you want to run and the privacy parameters to use.

1. The “Columns” box allows you to select which columns to query. Each column you specify will have an aggregation function applied and produce a differentially private result. You do *not* need to include columns that will be grouped-by. *To calculate the average age, specify the “age” column in this box.*

2. The “Grouping” box allows you to specify columns to group by. To calculate the average age grouped by city ID, specify the “city_id” column in this box.

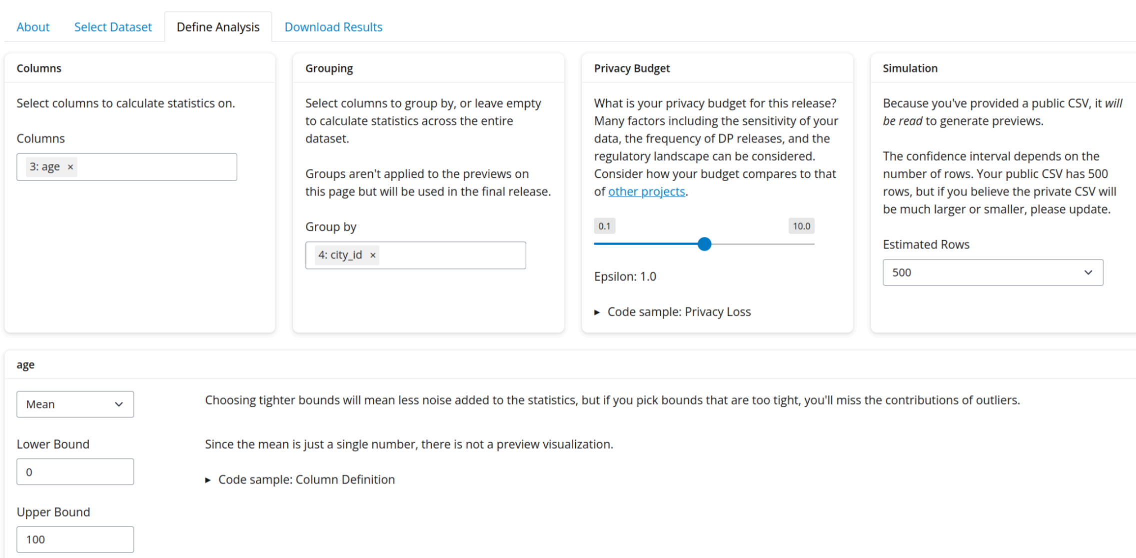
3. The “Privacy Budget” box allows you to specify the level of privacy protection - a smaller epsilon provides stronger privacy (but worse accuracy), and a larger one provides weaker privacy (but better accuracy). *For now, set epsilon to 1.0.*

4. The “Estimated Rows” box allows you to specify how many rows are in the dataset. *Since we uploaded a public dataset, the public dataset is used to determine the number of rows, and we can leave it as the default.*

5. The boxes below allow you to specify the aggregation function for each column you selected in step (1). Since we only selected a single column (age), only one aggregation function box will appear. DP Wizard supports three aggregation functions: Histogram (counting within intervals), mean, and median. *For our example, select the mean function.*

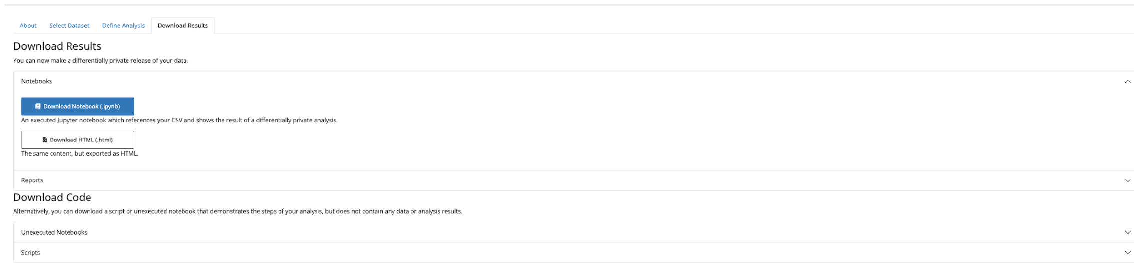
6. The “Lower Bound” and “Upper Bound” boxes in the mean aggregation function box allow you to set lower and upper bounds for individual data items in the private data. These are required because the differentially private version of the mean function first *clips* each data value to lie between the lower and upper bounds before computing the average. The best setting for the lower and upper bounds are highly data-dependent, and the results could be very inaccurate if the bounds are not set carefully. *Since our data includes ages, we can set the lower bound to 0 and the upper bound to 100.*

The settings should look like the screenshot below.



Step 4: Download Results

The “Download Results” tab allows you to download a Jupyter Notebook that actually computes the results of your specified query. In this tab, click “Download Notebook (.ipynb)” to download the notebook to your computer.



Step 5: Run the Notebook

Finally, we'll run the notebook produced by DP Wizard. For this tutorial, you can upload the notebook that you downloaded in step 4 to the provided Jupyter server and run it there - the server already has the required libraries installed.

1. Open the Notebook Server
2. Click Upload (top right) and select the notebook you downloaded in Step 4
3. Click "Run All Cells" from the "Run" menu to run all of the cells in the notebook

The results of your query appear near the bottom of the notebook. The cells above the results describe the query needed to compute the results. The results for our example appear as follows:

Query for `age`:

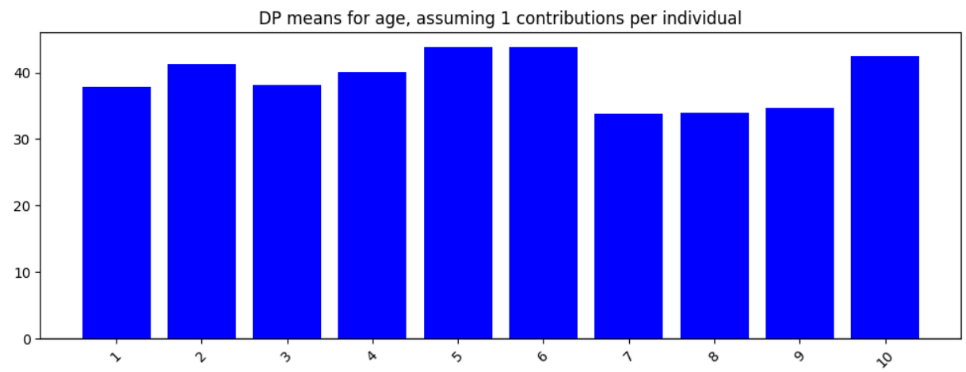
[7]:
```
groups = ["city_id"]
age_query = (
    context.query().group_by(groups).agg(age_expr)
    if groups
    else context.query().select(age_expr)
)
age_stats = age_query.release().collect()
age_stats
```

[7]: shape: (10, 2)

city_id	age
i64	f64
2	41.337066
5	43.882878
6	43.83044
9	34.751292
4	40.10243
10	42.425699
3	38.130961
7	33.840947
8	34.006445
1	37.785039

This table includes the differentially private results: noisy average ages for each city ID in the dataset. The notebook also graphs the results in the cell immediately below:

```
[8]: if groups:
    title = (
        f"DP means for age, "
        f"assuming {contributions} contributions per individual"
    )
    plot_bars(age_stats, error=0, cutoff=0, title=title)
```



The earlier parts of the notebook include important components for calculating the results. These chunks of code are programmatically generated by DP Wizard. Important sections include:

- “Expression for age”: this cell defines the actual query (the average) to be run on the selected column. It uses OpenDP’s “dp.mean” function to calculate the DP average.
- “Context”: this cell defines the privacy budget and CSV file to use for the private data.